Marathwada Mitramandal's

COLLEGE OF ENGINEERING, PUNE

An Autonomous Institute

Project Based Learning Project

Title of Project :Hotel Management System

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Abstract

The Hotel Management System is a modern web-based platform developed to simplify and automate the operations of hotel businesses. In traditional hotel management, processes such as room booking, customer registration, billing, and record maintenance are often handled manually, which can be time-consuming, error-prone, and inefficient. The proposed system overcomes these challenges by providing a centralized digital platform where all hotel-related information and services can be accessed anytime and from anywhere.

The system allows hotel administrators, staff, and customers to create accounts and manage various activities efficiently. It includes features such as room booking, check-in/check-out management, room availability status, customer details, billing, and payment tracking. Customers can easily browse available rooms, check prices, view amenities, and make reservations based on their preferences such as room type, budget, and duration of stay.

The website includes several important modules such as user registration, secure login, room management, booking system, billing system, customer feedback, and admin dashboard. The admin has full control over room allocation, customer records, bookings, payments, and report generation, ensuring smooth hotel operations and better decision-making.

The project is developed using modern technologies such as HTML, CSS, JavaScript, and Bootstrap for designing a responsive and user-friendly interface. Backend operations are handled using PHP or Node.js, and MySQL database is used to securely store customer details, booking information, and transaction records.

The Hotel Management System improves efficiency, reduces manual work, enhances customer satisfaction, and ensures better management of hotel resources. It provides convenience to both customers and hotel staff by making booking and management processes faster and more reliable. This project is also useful for students and businesses to understand real-world application development and database management systems.

In the future, the system can be enhanced with mobile applications, AI-based room recommendations, online payment gateways, chat support, automated notifications, and integration with third-party travel platforms. Overall, the project offers an efficient, user-friendly, and scalable solution for modern hotel management.

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CHAPTER 1

Introduction

1.1 Introduction to Hotel Industry

The hotel industry plays a vital role in the growth of the hospitality sector by providing accommodation, food, and various services to travelers and tourists. It includes different types of establishments such as luxury hotels, budget hotels, resorts, lodges, and guest houses. This industry generates employment opportunities in areas like management, housekeeping, catering, customer service, and tourism.

With the rapid growth of tourism and globalization, the demand for hotels and hospitality services has increased significantly. Customers now expect quick service, online booking , and better customer experience. Technology has transformed the hotel industry by introducing online reservation systems, digital payments, and automated management solutions, making operations more efficient and customer-friendly.

1.2 Introduction to Hotel Management System

A Hotel Management System is an online platform designed to manage hotel operations efficiently. It allows hotel staff and administrators to handle room bookings, customer details, billing, and other services in a systematic way. Customers can easily check room availability, view prices, explore amenities, and make reservations from anywhere using the internet.

The system provides detailed information about different types of rooms, pricing, availability status, and facilities offered by the hotel. It also allows users to filter rooms based on preferences such as budget, room type, and duration of stay. This system helps in reducing manual work and ensures quick and accurate management of hotel activities.

1.3 Motivation

The motivation behind developing this project is to overcome the limitations of traditional hotel management methods, which involve manual record keeping, phone-based bookings, and inefficient data handling.

Main motivations are:

1. To reduce manual work in hotel operations.
2. To save time in booking and managing rooms.
3. To provide all hotel-related information in one place.
4. To improve communication between customers and hotel staff.
5. To increase efficiency in managing bookings and payments.
6. To ensure accurate record keeping and reduce errors.
7. To provide quick access to room availability and booking status.
8. To enhance customer satisfaction through better service.
9. To make hotel services accessible anytime and anywhere.

1.4 Aim and Objective(s) of the Work

Aim: To develop a user-friendly website for listing and searching Hotels online.

Objectives:

To reduce manual work in hotel management processes.
To save time in room booking, check-in, and check-out operations.
To provide all hotel information (rooms, prices, services) in one place.
To improve communication between hotel staff and customers.
To increase efficiency in handling bookings, billing, and payments.
To maintain accurate customer and booking records.
To provide real-time updates on room availability and status.
To minimize errors in billing and data management.
To enhance customer satisfaction through faster and better services.
To make hotel services available online anytime and anywhere.

CHAPTER 2

LITERATURE SURVEY

Literature survey is the process of studying existing systems, research papers, websites, journals, and technologies related to the Hotel Management System. It helps in understanding how current hotel management solutions work, what features they provide, and what limitations exist in those systems. By analyzing previous work, better ideas and improvements can be implemented in the proposed project.

The hotel industry has evolved significantly with the advancement of internet technology. Earlier, hotel bookings and management were handled manually through phone calls, walk-ins, and paper records. This process was time-consuming, less efficient, and prone to errors. With the introduction of online hotel booking systems, customers can now search, compare, and book rooms anytime and from anywhere using computers or mobile devices.

Many existing hotel management systems provide features such as room booking, customer registration, billing, payment processing, and room availability tracking. These systems allow customers to view room details, prices, amenities, and make reservations easily. Hotel staff can efficiently manage bookings, check-ins, check-outs, and customer records using these digital platforms.

Research studies show that online hotel management systems improve operational efficiency, customer satisfaction, and service quality. Customers can access complete information before booking, and hotels can manage their resources effectively. Features like real-time room availability, online payments, and automated billing make the system faster and more reliable.

However, some existing systems still face common problems such as outdated information, lack of user-friendly interface, security issues, slow performance, and limited functionality. Some systems do not provide proper communication between customers and hotel staff or lack advanced features like notifications and feedback systems.

The proposed Hotel Management System is designed to overcome these limitations by providing a secure, responsive, and user-friendly platform. It offers features such as real-time booking, room management, customer records, billing system, admin control, and improved data handling. The system ensures better performance, accuracy, and convenience for both customers and hotel management.

STUDY OF LITERATURE SURVEY

Table 1.1

Sr No.	Paper Title	Publication & Year	Authors	Findings	Research Gaps
1	Online Hotel Booking Systems	IEEE, 2022	Various Authors	Improved online reservation system	Limited personalization features
2	Web-Based Hotel Management Systems	IJERT, 2021	Various Authors	Better data management	Weak security modules
3	Smart Hotel Recommendation System	Springer, 2023	Various Authors	AI-based suggestions	High implementation cost

CHAPTER 3

Methodology & Algorithms Used

3.1 Methodology

The Hotel Management System is developed using a structured software development process to ensure proper planning, design, coding, testing, and deployment. Each phase plays an important role in building an efficient and user-friendly system.

1. Requirement Gathering

In this phase, all system requirements are collected from hotel staff, administrators, and customers. Features such as user registration, login, room booking, check-in/check-out, billing system, room availability, and admin dashboard are identified..

2. System Design

The layout of the website, navigation flow, and database design are prepared. Wireframes and interface designs are created to ensure smooth user experience and easy navigation.

3. Database Design

Database tables are created for users, rooms, bookings, payments, and admin. Relationships between tables are established to ensure proper data storage and efficient management.

4. Frontend Development

The user interface is developed using HTML, CSS, JavaScript, and Bootstrap to create responsive and attractive web pages for customers and admin.

5. Backend Development

Server-side programming is done using PHP or Node.js. It handles user authentication, booking management, payment processing, and database connectivity.

6. Testing and Deployment

The system is tested for bugs, errors, security issues, and performance. After successful testing, the system is deployed on a local or online server.

3.2 Formula Used

Room Billing Formula

Total Bill=Room Price Per Day×Number of Days **Room Price Per Day** = Cost of room per day

This formula helps calculate the total amount to be paid by the customer based on their stay duration.

Search Result Percentage Formula

Match Percentage=Total RoomsMatching Rooms×100

This helps analyze how many rooms match user search criteria

3.3 Example

Example 1: Room Billing

If room price per day is ₹2000 and number of days is 5:

Total Bill = 2000 × 5 = ₹10,000

So, total bill = **₹10,000**

3.4 Algorithms

Algorithm 1: User Registration

1. Start
2. Enter user details
3. Validate input data
4. Store data in database
5. Display success message
6. Stop

Algorithm 2: room Search

1. Start
2. User enters location, budget, type
3. System checks database
4. If property found, display results
5. Else show “No room Found”
6. Stop

Algorithm 3: Room booking

1. Start
2. User selects room
3. Fill booking details
4. Check room avilibility
5. If available, confirm booking
6. Update database
7.Display booking confirmation
8.Stop

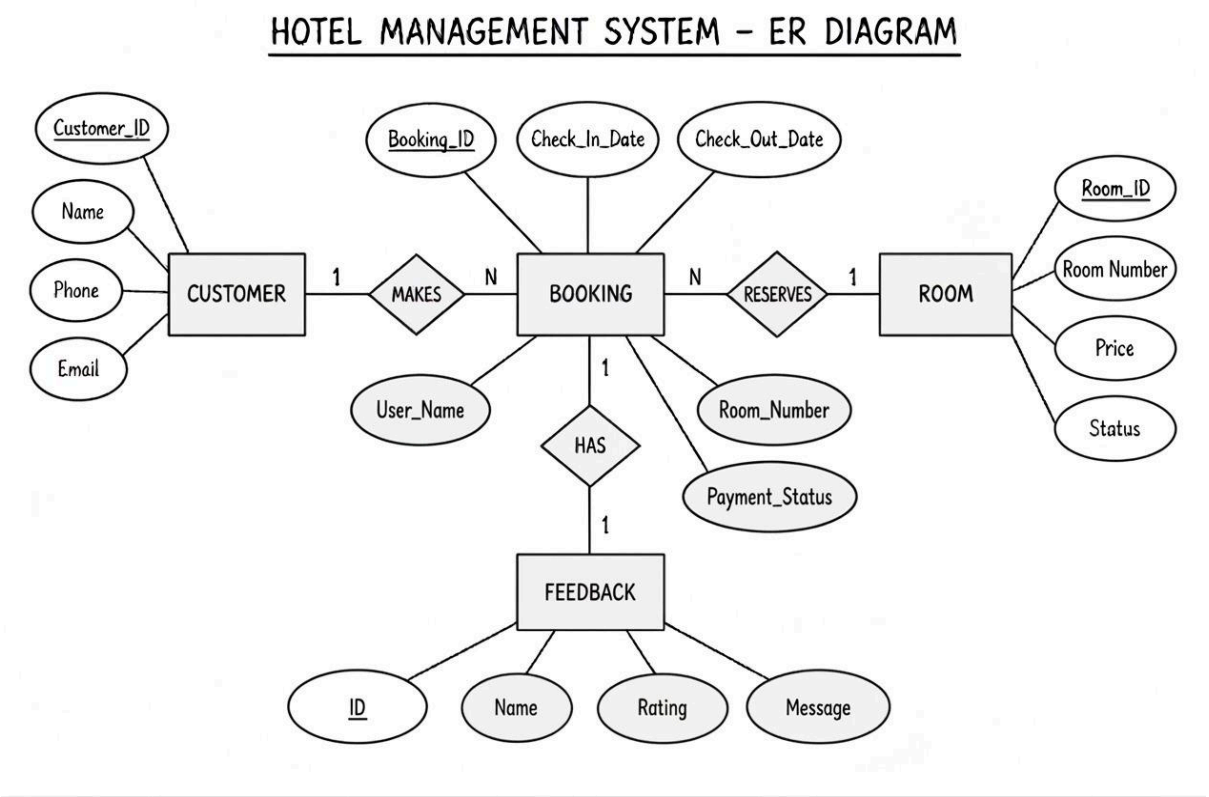
Algorithm 4: Admin Management

1.Start
2.Admin login
3.View customers and room details
4.Add / Update / Delete room records
5.Manage bookings and payments
6.Generate reports
7.Logout
8.Stop

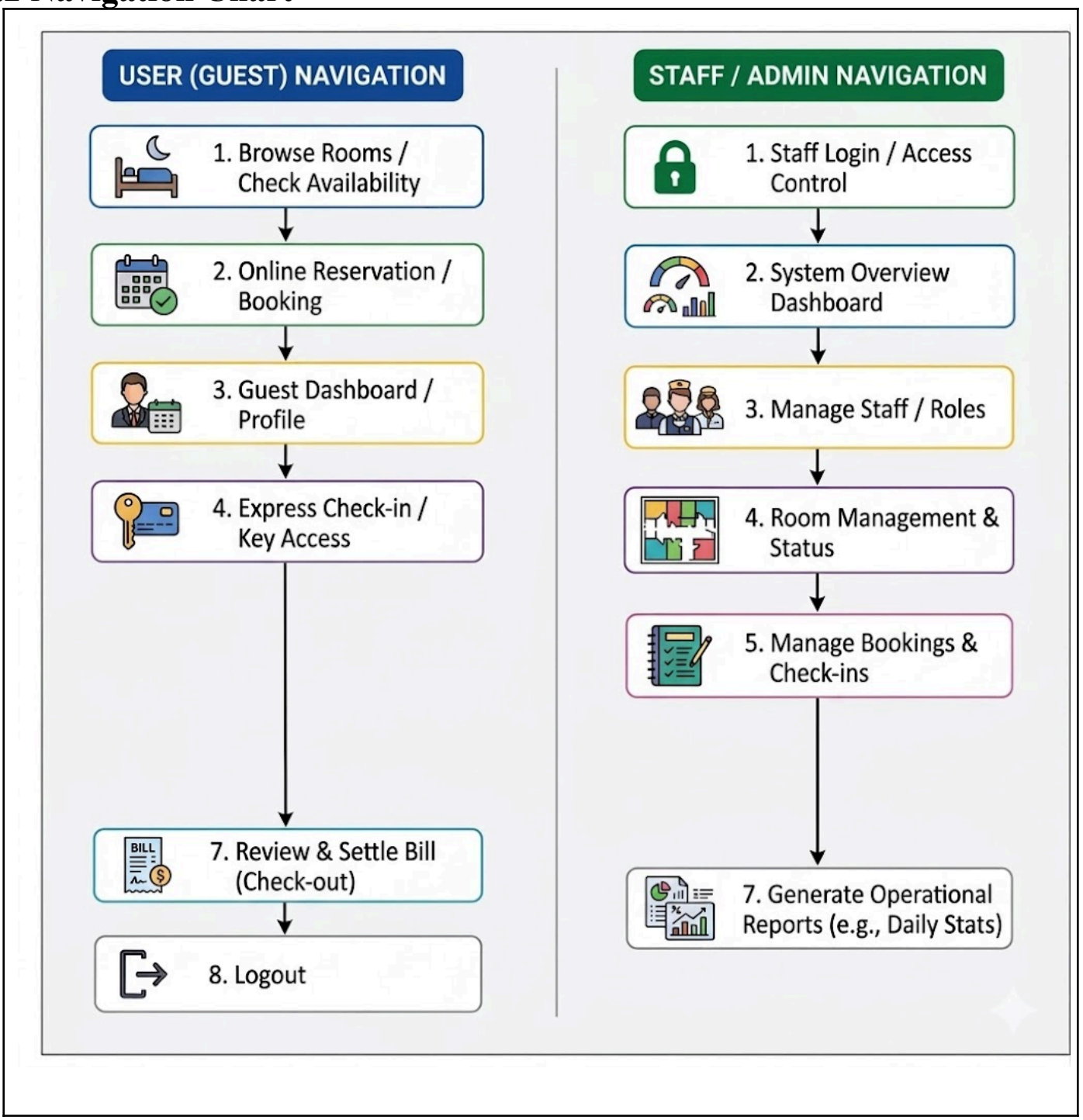
CHAPTER 4

ER Diagrams And Navigation Chart

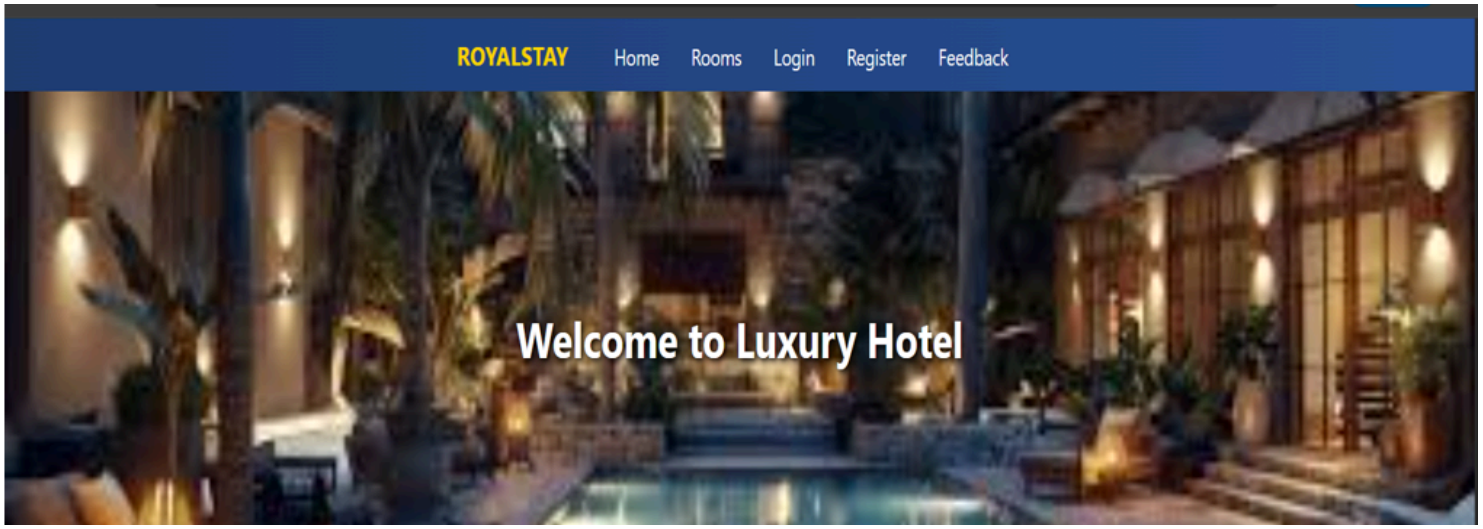
4.1 ER Diagrams

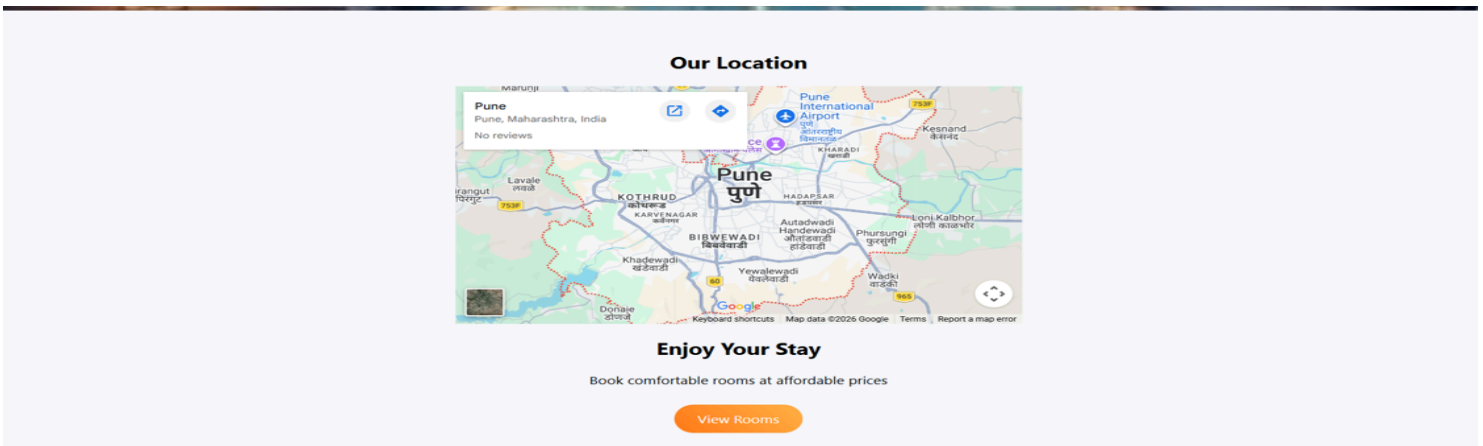


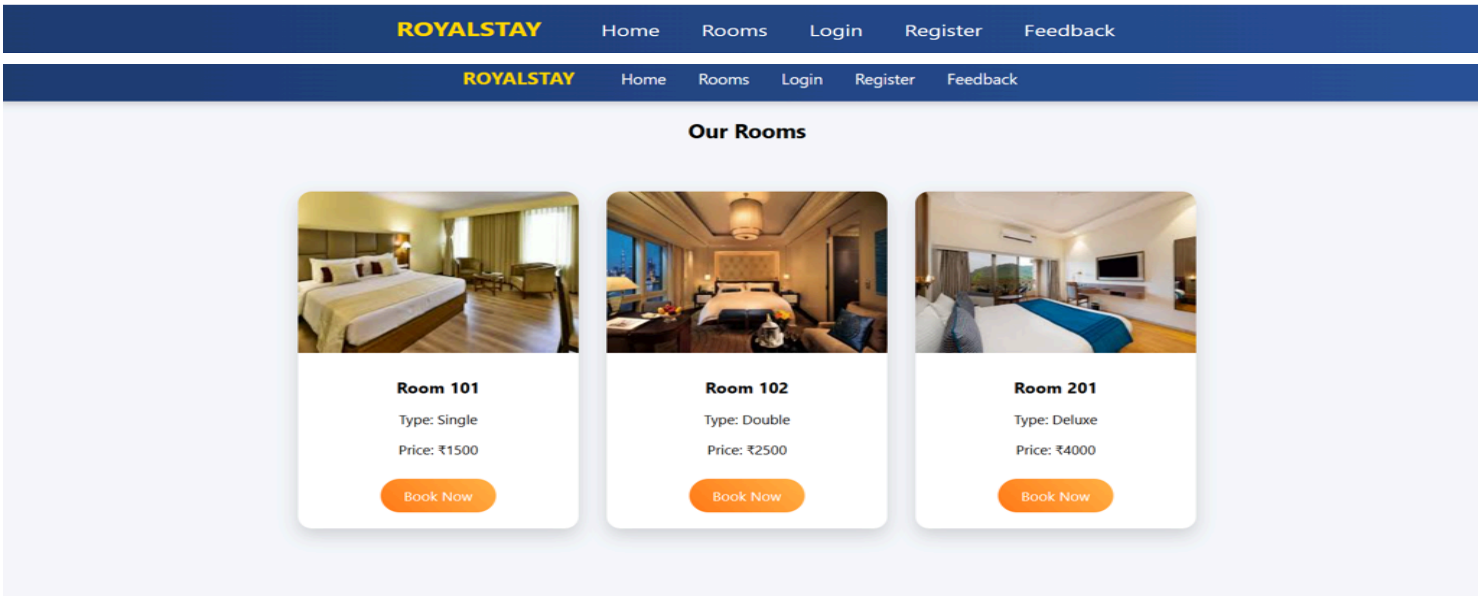
4.2 Navigation Chart



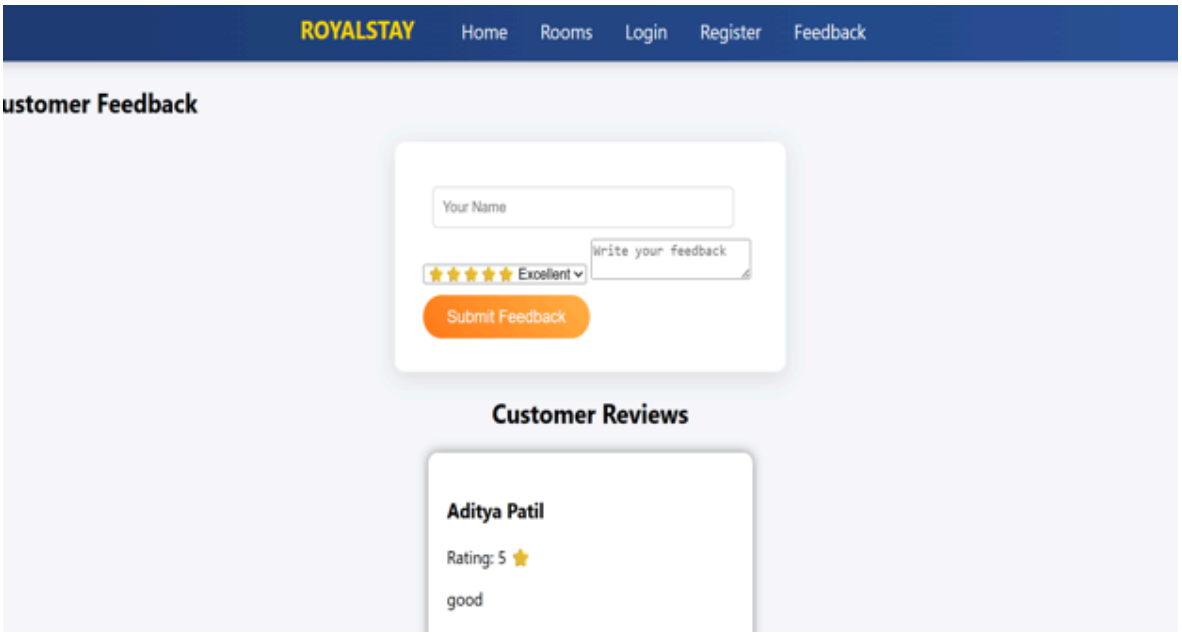
4.1 Screenshots and Github Link :







ROYALSTAY
Home
Rooms
Login
Register
Feedback
Welcome to ROYALSTAY
Experience luxury and comfort with our premium rooms
Explore Rooms
Book Now
3
Total Rooms
5
Total Bookings
4
Registered Users


ROYALSTAY
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Register
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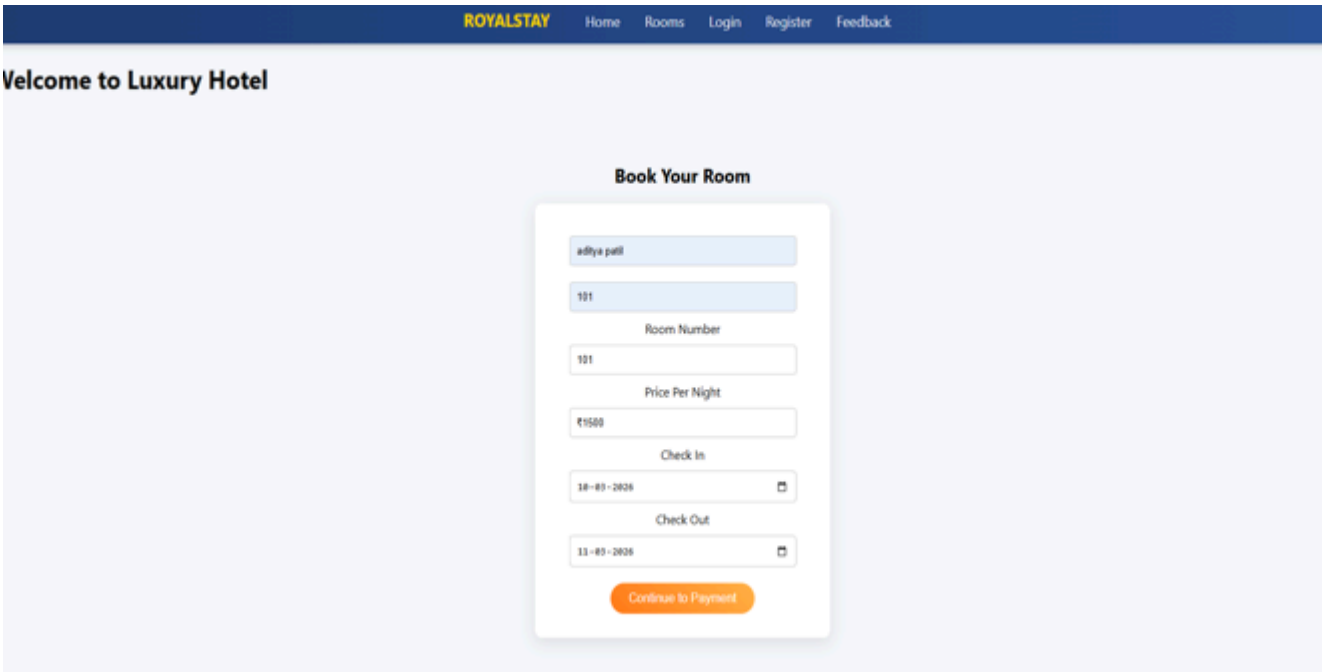
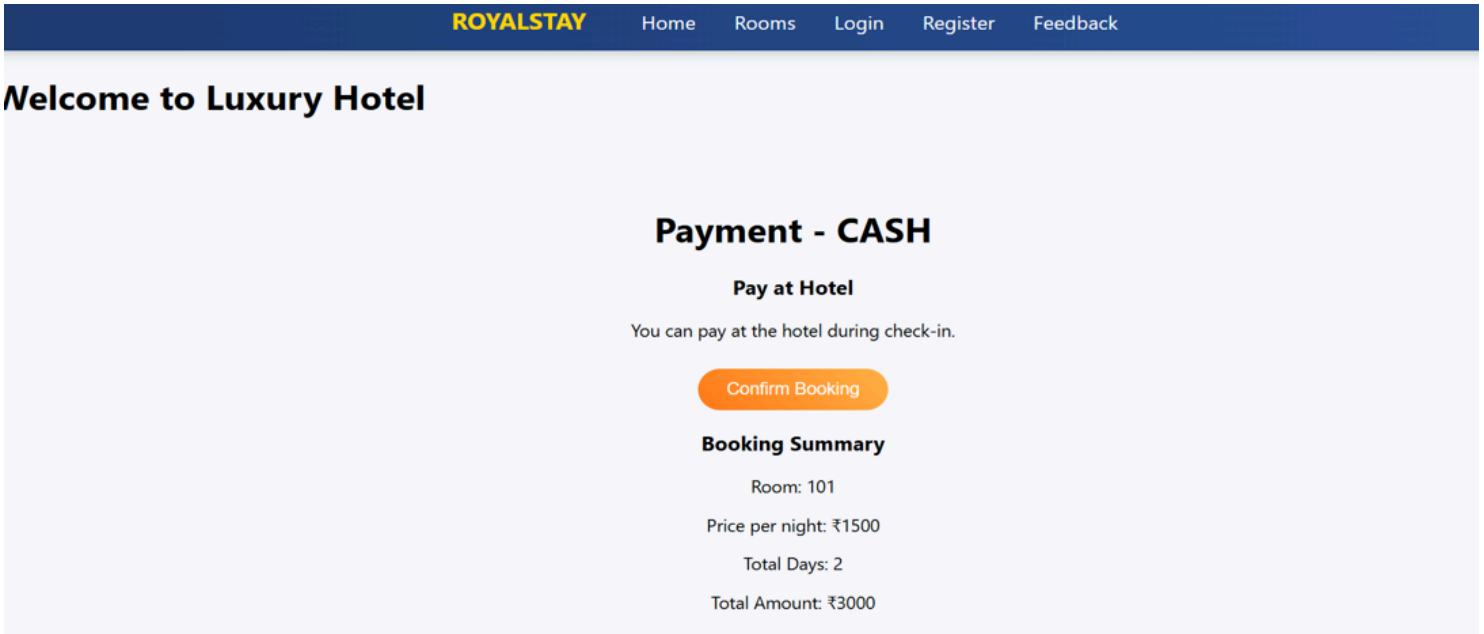
Your Name
Write your feedback
Excellent
Submit Feedback
Customer Reviews
Aditya Patil
Rating: 5
good


ROYALSTAY
Home
Rooms
Login
Register
Feedback
Velcome to Luxury Hotel
Book Your Room
aditya patil
101
Room Number
101
Price Per Night
₹1500
Check In
10-03-2026
Check Out
11-03-2026
Continue to Payment



GITHUB LINK:

[ROYAL STAY]

CHAPTER 5
Outline and Future Scope

5.1 Outline

The Hotel Management System is designed to simplify hotel operations such as room booking, customer management, and billing through an online platform. It provides users with easy access to room information, prices, availability status, and hotel services. The system includes modules such as user registration, login, room management, booking system, billing system, and admin dashboard. It helps reduce manual effort, improves efficiency, and saves time for both customers and hotel staff.

5.2 Future Scope

The project can be further improved by adding advanced features in the future:

1. **Mobile Application Development**
   Android and iOS apps can be developed for better accessibility.
2. **AI-Based Room Recommendation**
   Smart suggestions based on user preferences and booking history.
3. **Online Payment System**
   Integration of secure payment gateways for easy transactions.
4. **Automated Notifications**
   SMS/Email alerts for booking confirmation, check-in, and reminders.
5. **Smart Room Services Integration**
   Integration with IoT devices for smart room control (lights, AC, etc.).
6. **Chat System**
   Live chat between customers and hotel staff.
7. **Multi-Language Support**
   Support for multiple languages to reach more users.
8. **Advanced Security Features**
   OTP login, data encryption, and secure payment processing.

5.3 Benefits in Future

1.Faster and efficient hotel operations
2.Improved customer experience
3.Better management of bookings and records
4.Increased accuracy and reduced errors
5.Enhanced communication between staff and customers
6.Wider reach for hotel services
7.Smart and automated hotel management system

CHAPTER 6

Conclusion

The Hotel Management System is a smart and efficient solution for modern hotel operations such as room booking, customer management, and billing. It provides a simple online platform where users can check room availability, view details, make reservations, and manage their bookings easily. This reduces the traditional manual process and saves valuable time and effort.

The system is beneficial for customers, hotel staff, and administrators. Customers can easily find suitable rooms and book them according to their preferences, while hotel staff can efficiently manage bookings, check-ins, check-outs, and payments. The admin can manage customer records, room details, and overall system operations effectively. Secure login, organized database management, and a user-friendly interface make the system reliable and easy to use.

This project is also helpful for students and businesses to understand real-world hotel management systems. It demonstrates the practical use of web technologies such as frontend design, backend development, and database connectivity.

Overall, the Hotel Management System improves efficiency, accuracy, and customer satisfaction in the hospitality sector. With future enhancements like mobile applications, AI-based recommendations, online payment integration, and automated notifications, the system can become more advanced and valuable

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